Sachin S Nair

AI & ML · Software · UI/UX

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PROFILE

Computer Science student (AI & ML) at College of Engineering Chengannur, graduating in 2028. I build machine learning projects end to end: training models in Python and putting them behind a working interface. I also design interfaces in Figma.

PROJECTS

StockSense AI — Stock price forecasting web app

- Built a 3-layer stacked LSTM in Keras/TensorFlow that forecasts N-day stock prices from Yahoo Finance data, with confidence intervals.
- Evaluated the model with RMSE, MAE and MAPE, and wrapped it in a Streamlit dashboard with Plotly candlestick charts, moving averages and CSV export of forecasts.

Tech: Python, Keras, TensorFlow, scikit-learn, Streamlit, Plotly, yfinance · GitHub

Retinal AI — Diabetic retinopathy detection and vessel segmentation (team of 3)

- Developed the ResNet-50 classifier in PyTorch that grades retinal fundus images into five severity levels and reports class probabilities and confidence.
- Input pipeline uses CLAHE contrast enhancement, 224×224 resizing and ImageNet normalisation.
- Team project: the full app also includes U-Net vessel segmentation and YOLO lesion detection behind a FastAPI backend and React interface, with downloadable PDF reports.

Tech: Python, PyTorch, OpenCV, ResNet-50 · GitHub

EDUCATION

B.Tech, Computer Science and Engineering (AI & ML) 2024 – 2028 (expected)
College of Engineering Chengannur

Class XII, CBSE 2024
Good Shepherd Public School & Junior College, Thengana, Changanassery, Kottayam

Class X, CBSE 2022
Mar Ivanios Bethany Senior Secondary School, Kalayapuram, Kollam

SKILLS

Languages
Python (primary), C, C++, Java

Machine learning
scikit-learn, TensorFlow, Keras, PyTorch, OpenCV, NumPy, Pandas

Tools
Git, GitHub, VS Code, Streamlit, Gradio, Hugging Face

Design
Figma, Canva, Adobe (UI/UX)

LANGUAGES

Malayalam (native)
English (proficient)

INTERESTS

Photography, UI/UX design